

Computational Fluid Dynamics for Aerospace Applications

Lecture 1 Introduction

Introduction

- Why use CFD?
- What is CFD?
- How does CFD get used?
- Syllabus

Why use CFD?

- Why don't we have a reusable commercial hypersonic aircraft that can transport humans?



Generated in Gemini

Why use CFD?

- Complex physics – Shockwaves, Thousands of Kelvin temperatures at surfaces, Complex air chemistry, etc.

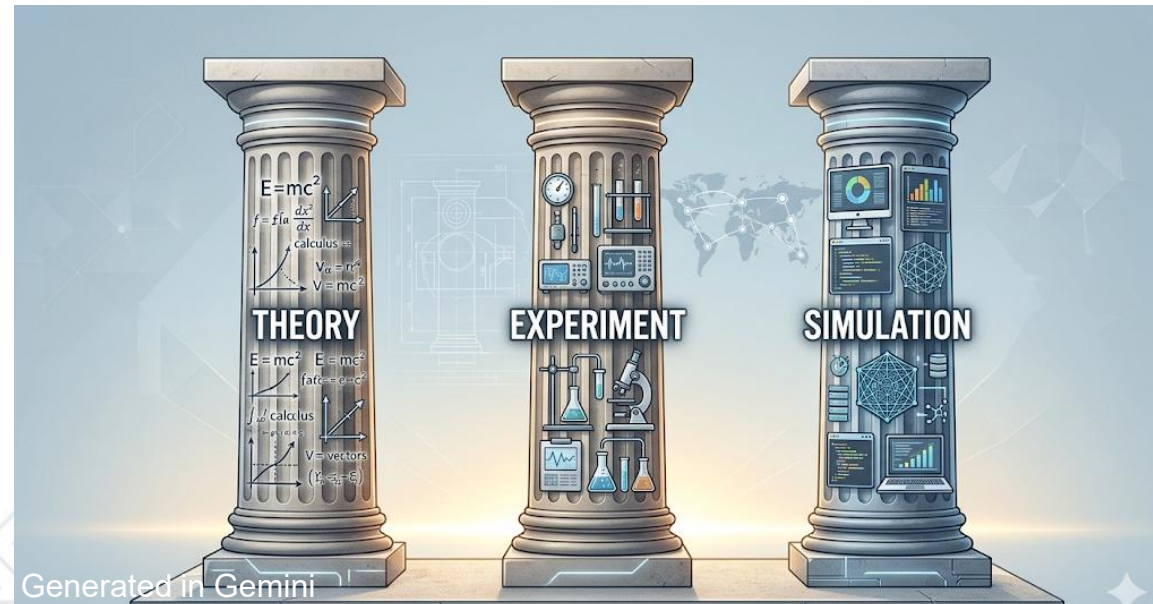


Why use CFD?

- How do we design for these conditions?
 - Theory is severely limited for complex geometries and physics
 - Experiments give us the right answer but
 - No wind tunnels of this size exist for these conditions
 - Maybe a plasma torch or arcjet gives close enough temperature?
 - No shockwaves or other flow features
 - How do you measure flows moving at thousands of meters per second?
 - How do you measure temperatures greater than 3000 K?
- Theory and Experiments cannot always study our system
- Experiments could be costly or infrastructure limited

Why use CFD?

- Simulations are increasingly becoming a third engineering tool
 - Allow for prediction of complex flow systems
 - Relatively cheap and fast
 - Probe physics everywhere at all times
 - Share a common infrastructure (compute clusters)
 - Accuracy can be an open question



Why use CFD?

- We use CFD because it can help us design engineering systems
 - Faster
 - At lower cost
 - For conditions we cannot otherwise test/predict
- CFD doesn't replace theory or experiments, it complements them
 - The engineering design process often utilizes all three

What is CFD?

- Computational Fluid Dynamics (CFD) is a branch of fluid mechanics that uses numerical analysis and data structures to analyze and solve problems that involve flows – Wikipedia
- CFD consists of computing multidimensional flows governed by the Navier-Stokes or Euler equations with the goal of obtaining both physical insight into the flow dynamics as well as quantitative predictions of engineering flow systems – Gretar Tryggvason

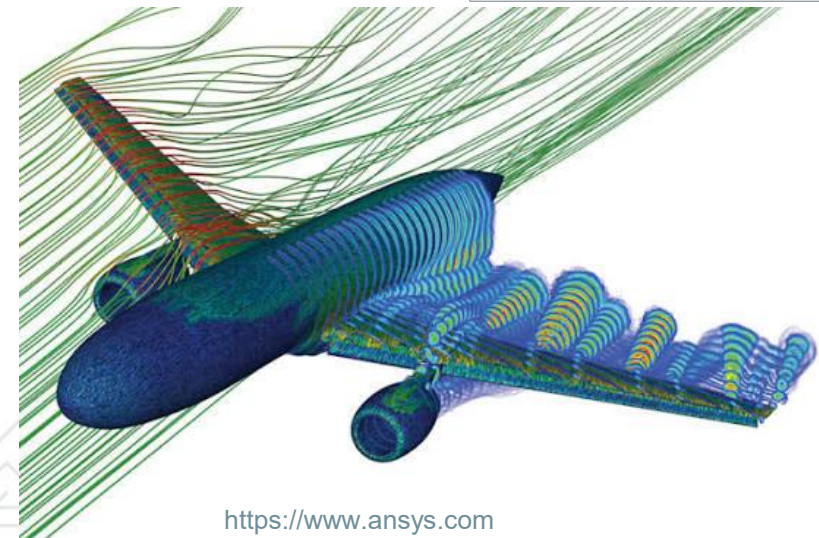
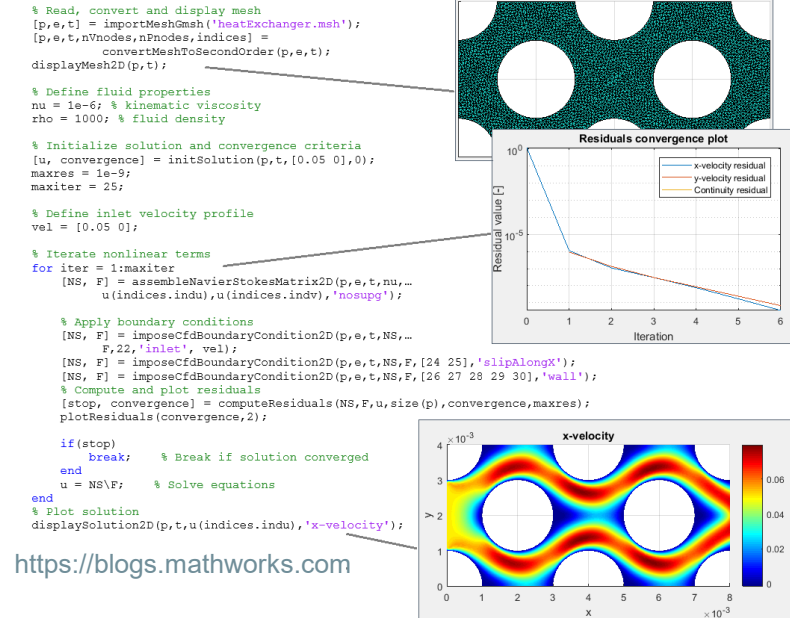
What is CFD?

We are going to:

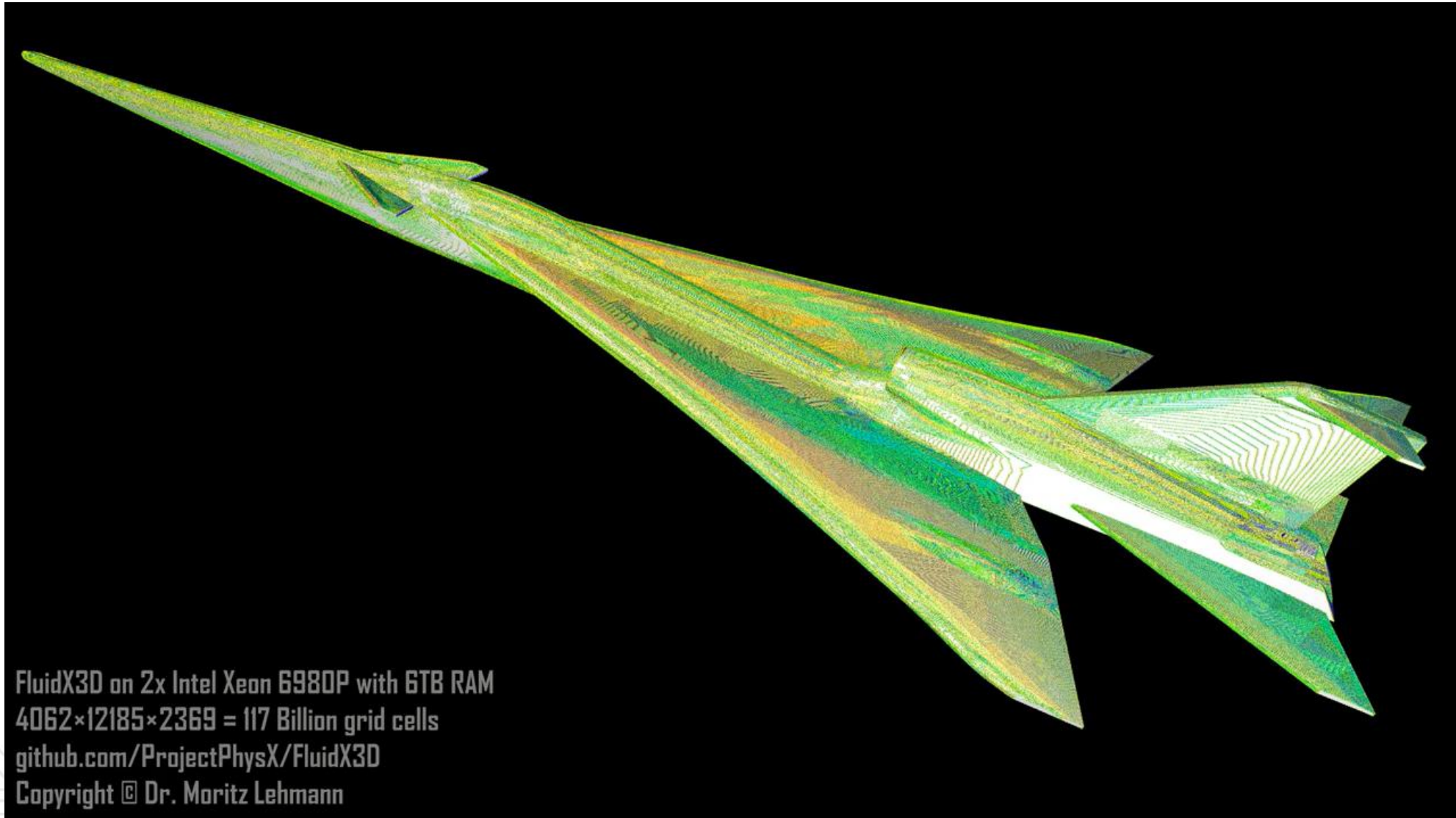
- 1) approximate the Navier-Stokes equations into an algebraic form (easy to calculate)
- 2) solve them on a massive number of points and times (to minimize our error)
- 3) and analyze the massive amount of resulting data

Two broad types of CFD users

- Development
 - Creating better CFD codes (faster, cheaper, more accurate)
 - Implementing new physical models to study new types of problems
 - Code + Math with simple configurations
- Engineering
 - Predicting the behavior of systems to design better products/devices
 - Academic exploration of physics in novel conditions
 - Complex, expensive configurations



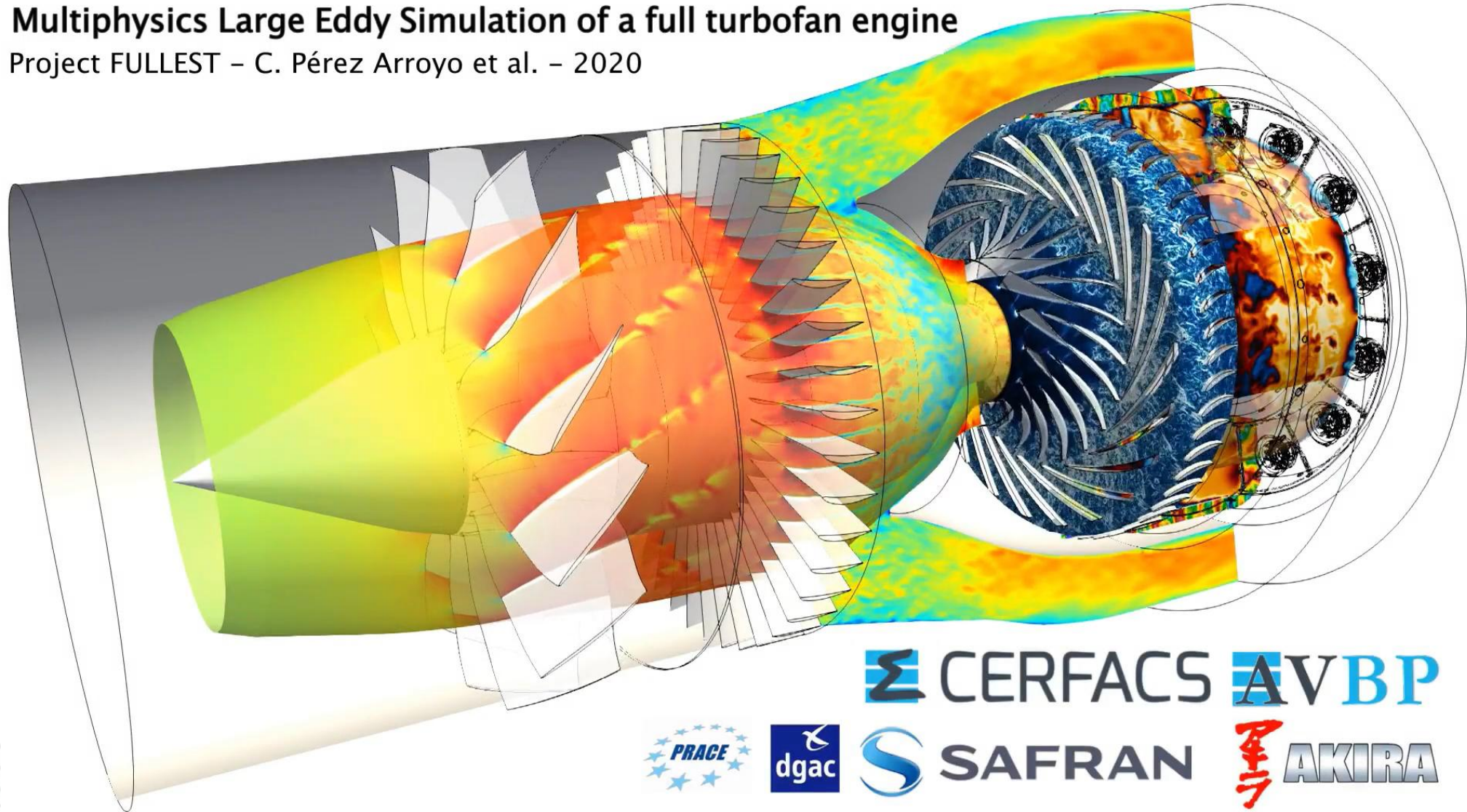
How does CFD get used?



How does CFD get used?

Multiphysics Large Eddy Simulation of a full turbofan engine

Project FULLEST – C. Pérez Arroyo et al. – 2020



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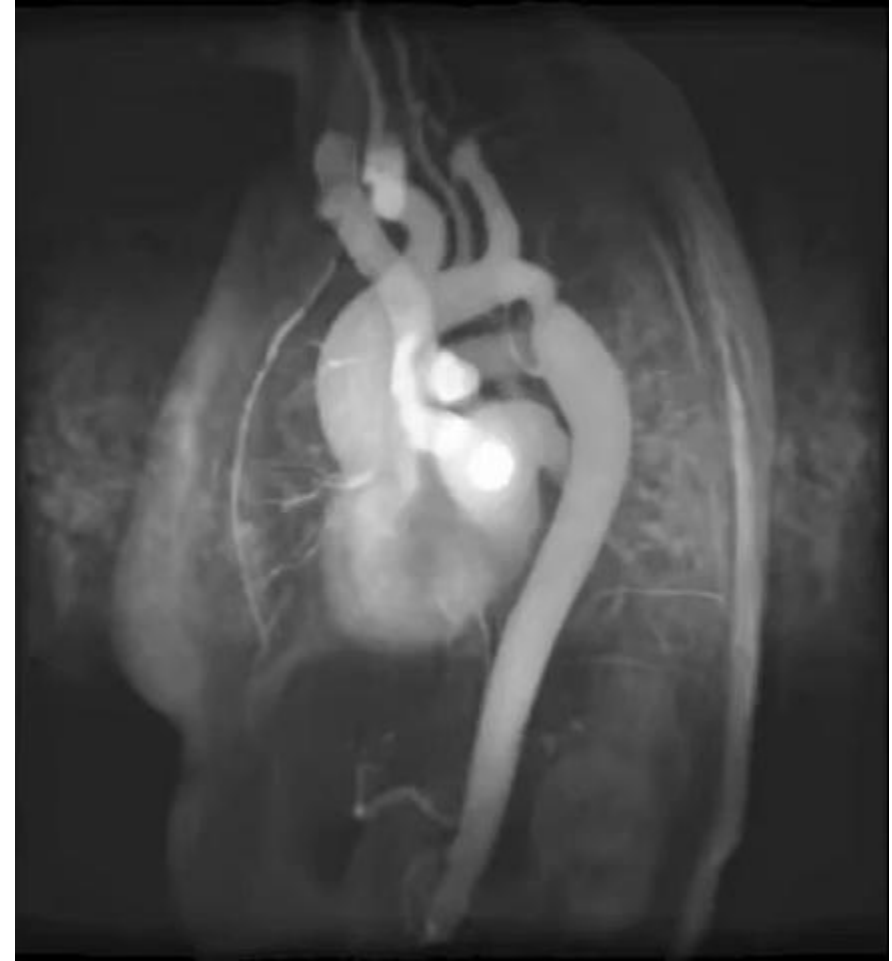
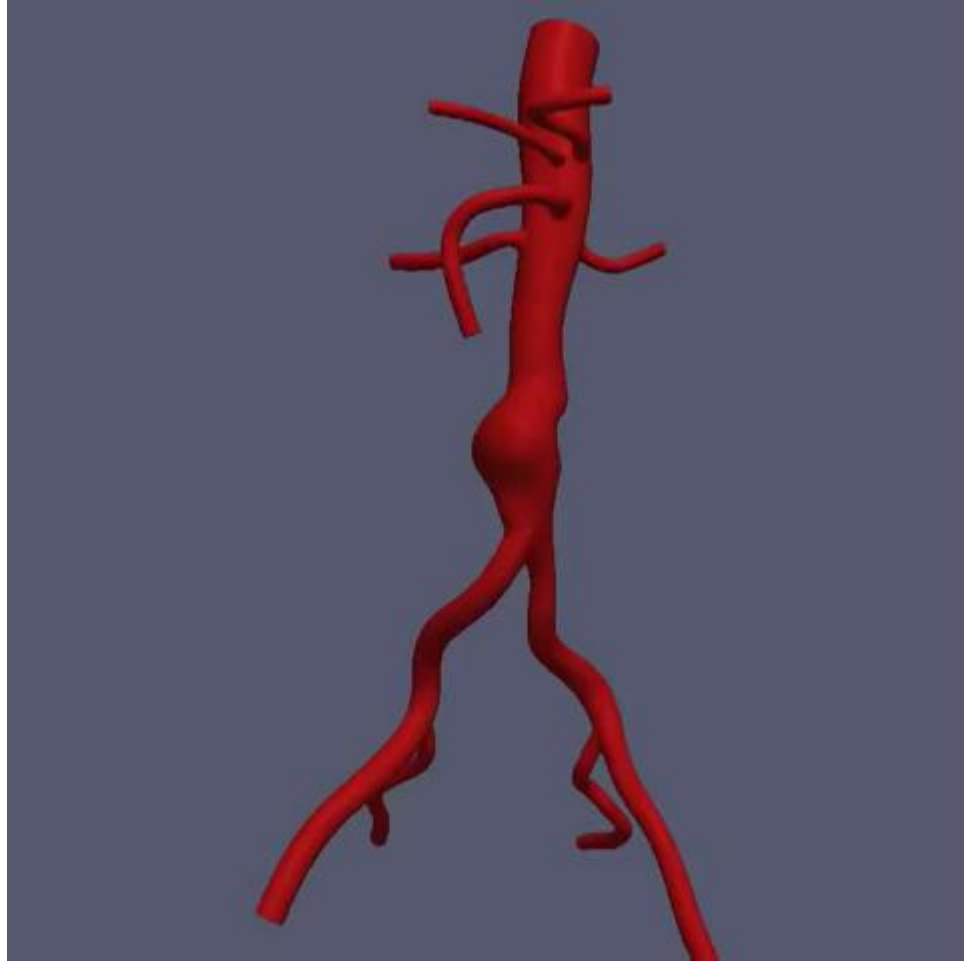
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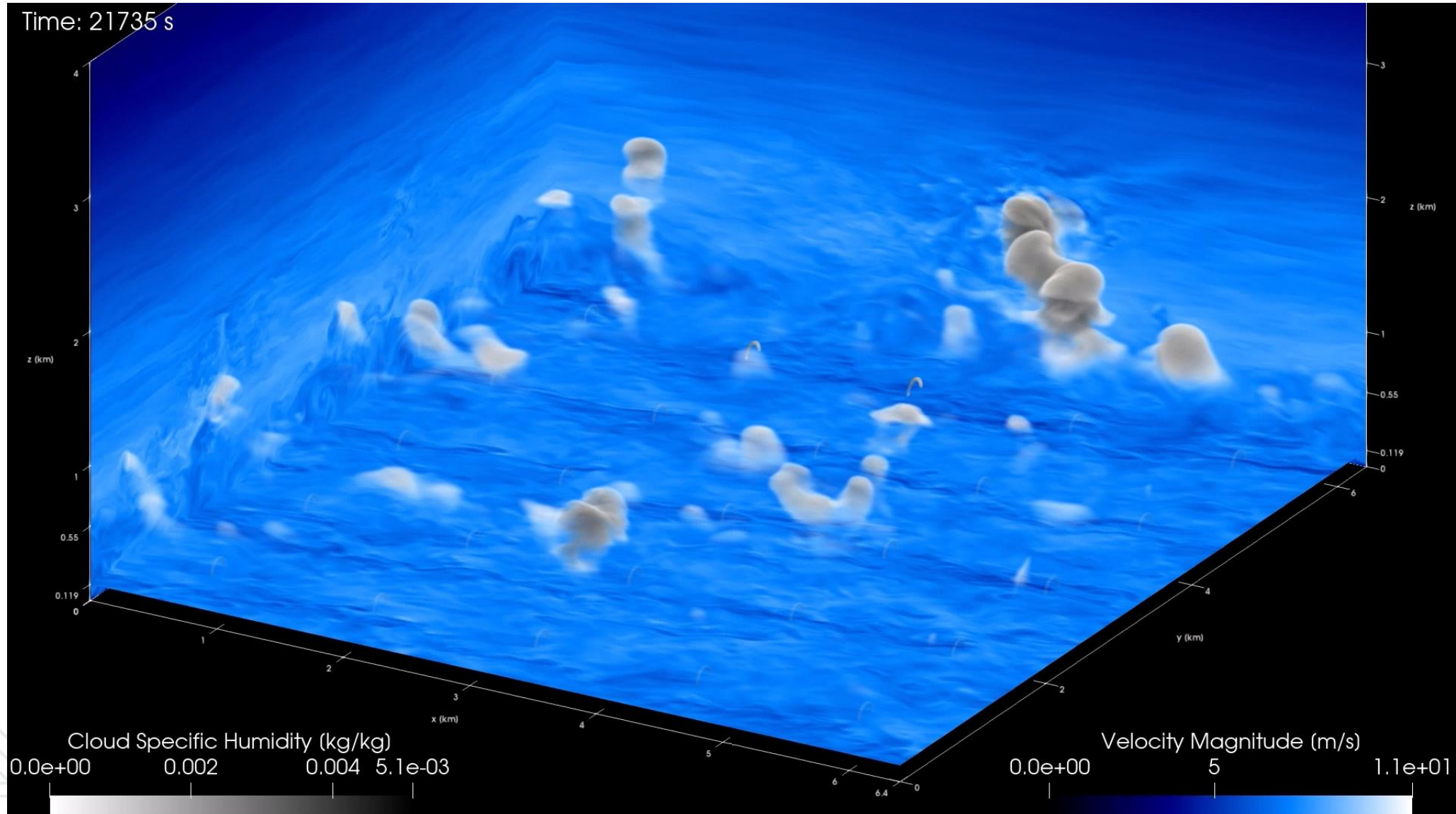
How does CFD get used?

Prof. Arzani – Blood flow



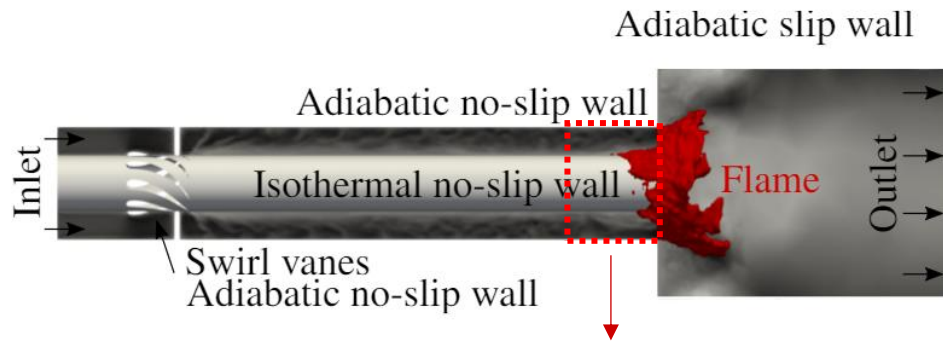
How does CFD get used?

Prof. Calaf – Environmental flows

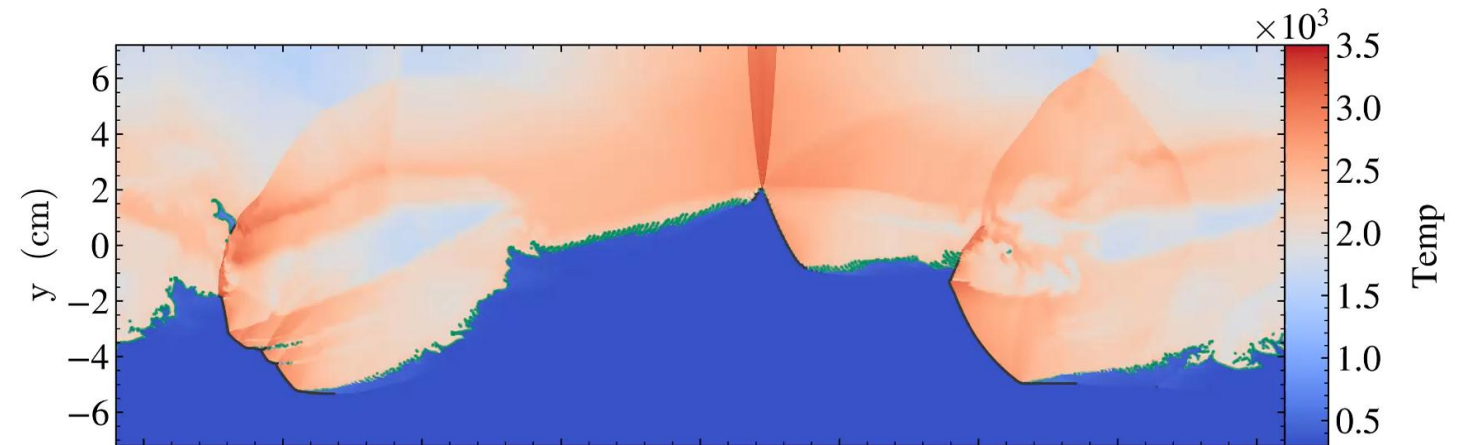
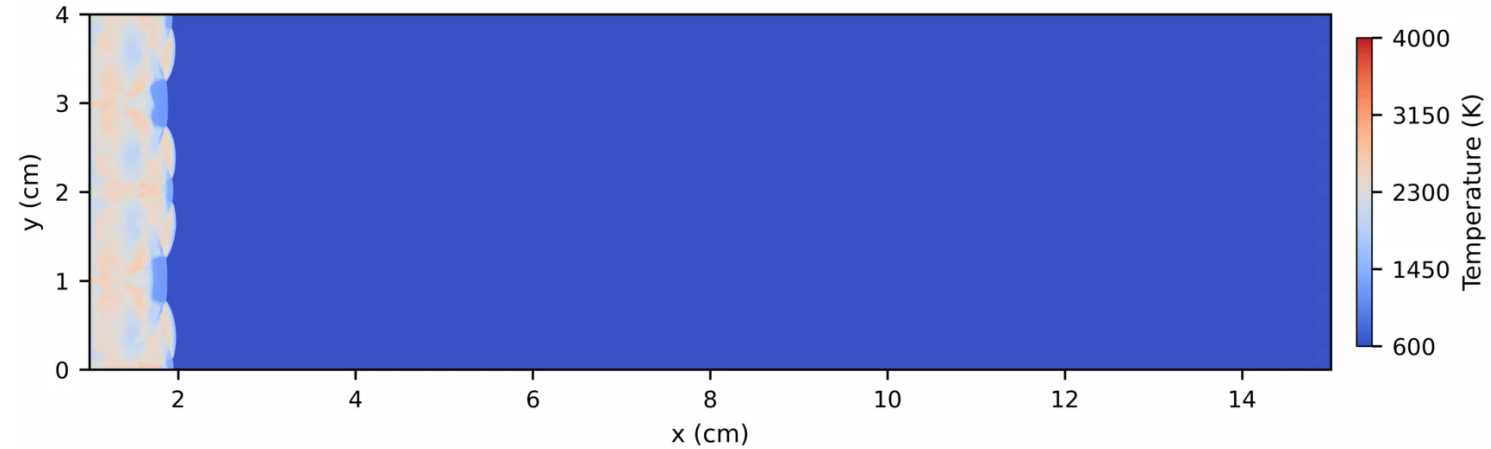


How does CFD get used?

Prof. Novoselov – Combustion



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Syllabus

Computational Fluid Dynamics for Aerospace Applications ME EN 5960/6960-002

Monday/Wednesday 11:50am-1:10pm

WEB L122

Fall 2026

Syllabus and course policies are subject to revision. Newest version available on Canvas

Instructor: Alex Novoselov

Office: MEK 1342

Email: alex.novoselov@utah.edu

Office Hours: TBD

Textbooks: This course is based on the below textbook. Students are not required to purchase any textbooks as all necessary content will be delivered through classroom lectures. If interested in learning more, students may seek out the textbook.

[Computational Fluid Dynamics: The Basics with Applications, John D. Anderson, Jr.](#)

Course Website: The course website is on Canvas and will include announcements, lecture materials, and assignments. It is your responsibility to check Canvas for messages, assignments, and updates.

Email: All emails from students to the instructor and teaching assistant should be sent to the address listed above. You can expect a response within one business day. Emails should only be sent from utah.edu addresses to guarantee receipt.

Syllabus

Prerequisites: 'C' or better in ME EN 3650.

Course Description: This course serves as an introduction to Computational Fluid Dynamics (CFD) with a particular emphasis on aerospace applications. In the first part of the course, students will learn the fundamental theory underlying CFD including the governing equations of fluid mechanics, numerical approaches to solving partial differential equations while maintaining stability, and algorithms for solving compressible flows. In the second part of the course, students will apply this knowledge to generate meshes and perform simulations of various aerospace applications in commercial CFD software commonly used in industry. Additionally, students will learn to utilize analysis and visualization techniques that allow them to interpret their CFD results.

Course Objectives: Students completing this course will:

- 1) Learn the fundamental governing equations relevant to CFD.
- 2) Utilize numerical approximations to solve partial differential equations.
- 3) Configure simulations to maintain numerical stability.
- 4) Apply traditional algorithms for solving compressible flows to the governing fluid equations.
- 5) Become familiar with a commercial CFD software package used in the aerospace industry.
- 6) Generate a computational mesh of a geometry.
- 7) Simulate aerospace systems using commercial CFD software.
- 8) Perform analysis and visualization of simulation results using common tools and software.

Syllabus

Grading Policy: The grading distribution for this course is given below:

ME EN 5960/6960-002	
Assignments	20%
Project #1	25%
Project #2	25%
Final Project	30%

Grading Scale: At the instructor's discretion, final grades may be curved upwards based on overall class performance. This can only raise students' final grades.

Grades	Points	Explanation
A	92.5 - 100	Excellent performance, superior achievement
A-	89.5 - 92.4	
B+	86.5 - 89.4	Good performance, substantial achievement
B	82.5 - 86.4	
B-	79.5 - 82.4	
C+	76.5 - 79.4	Standard performance and achievement
C	72.5 - 76.4	
C-	69.5 - 72.4	
D+	66.5 - 69.4	Substandard performance, marginal achievement
D	62.5 - 66.4	
D-	59.5 - 62.4	
E	0 - 59.4	Unsatisfactory performance and achievement

Syllabus

Project #1

The first project will require students to use what they learn in the first part of the course to write their own CFD solver. The exact code specifications and the system to be simulated will be provided to students. Both the code and a report describing the code and its output results will be required and graded.

Project #2

The second project will require students to use what they learn in the second part of the course to simulate a specified aerospace system using a commercial CFD code. A report describing the meshing, simulation, and data analysis will be required.

Final Project

Students will be tasked with choosing their own project of interest. This may be developing new features in their own code or simulating a new system using a commercial code. Students may work in groups for the final project. Project topics must receive approval from Prof. Novoselov ahead of time. A report documenting the project will be graded.

Assignments

A small number of assignments will be administered between projects to assist students with developing the skills necessary for completing the projects. Components of these assignments may be used directly in projects where applicable.

Student Expectations: Students are expected to come to all lectures. Lecture notes are uploaded to Canvas before class, with derivations and problem solutions left blank. Students are expected to take notes during lectures. Derivations and problems, as solved in class, will be uploaded after class. The textbook primarily serves as a tool for further education on topics students wish to learn more about or do not fully comprehend during lecture.

Syllabus

Academic Misconduct

- Your goal in this class should be to learn. Generally, academic misconduct aligns with things that keep you from learning. If your intention is to learn, academic misconduct shouldn't be an issue.
- You may (and should!) work together on projects and assignments, but anything submitted must be your own work.
- Leverage tools in this class (e.g., AI) but stick to what each assignment explicitly allows you to use. If unsure, ask!